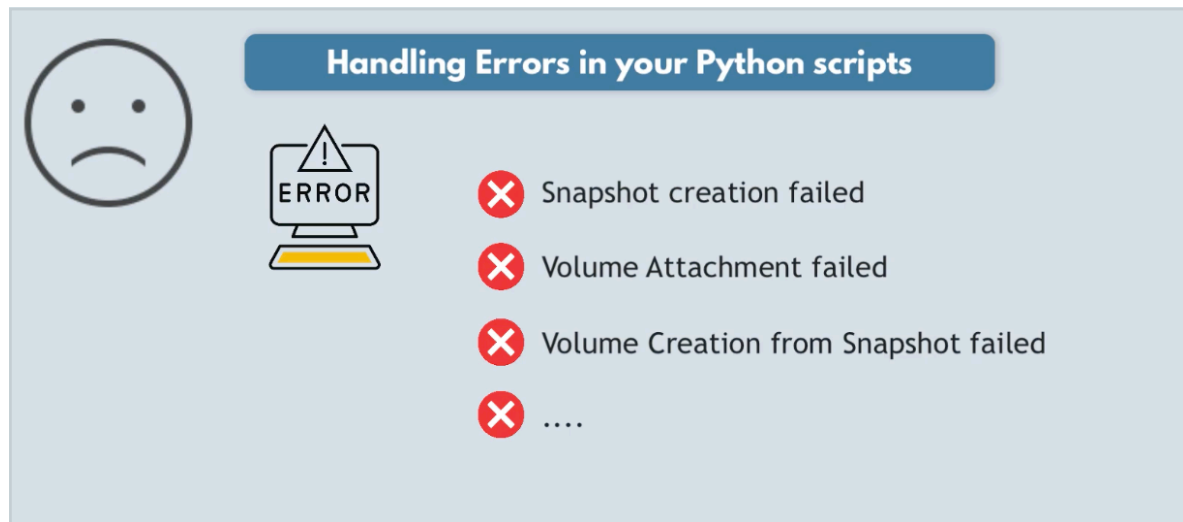


Module 14 - chapter 12

Handling Errors

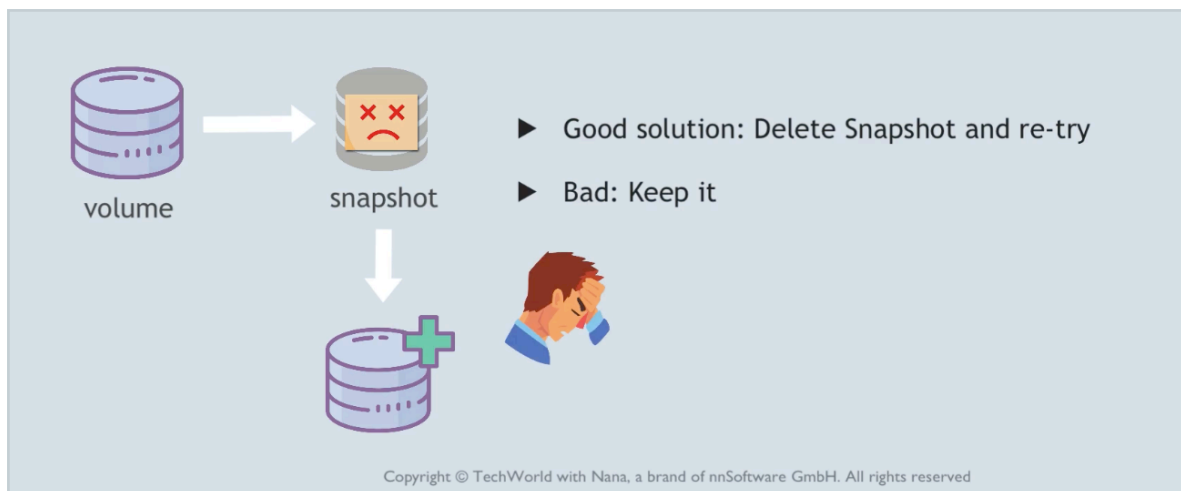
Many things can go wrong when running the python script



So we need to handle the possible errors and for that we also write code.

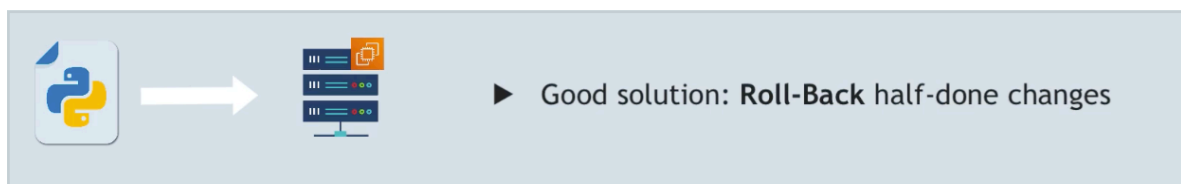


If a snapshot creation goes wrong the best thing is delete the snapshot and re-try

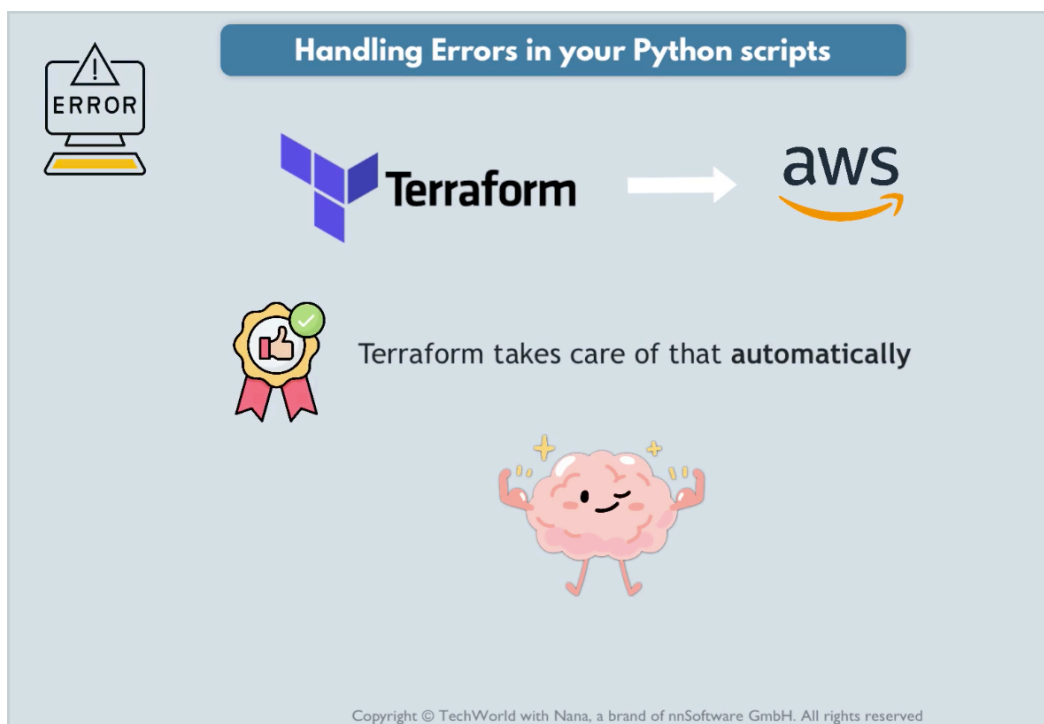


We don't want to keep a snapshot that can be used to restore a volume if the snapshot is already wrong.

Another example is that we update something in the server and some of the changes work but others don't. Then it's better to roll back.



Terraform can handle this kind of situations thanks to the STATE.



But python doesn't have a STATE so we need to take care of errors by

ourselves



Python **doesn't** have a state, so we need to take care by ourselves

The syntax for that is TRY/EXCEPT

So let's improve this code:

```
volume-backups.py x
1 import boto3
2
3
4 ec2_client = boto3.client(*args: 'ec2', region_name='eu-west-2')
5
6 def create_volume_snapshots(): 1 usage Astrid Caballero-Pardo
7     volumes = ec2_client.describe_volumes(
8         Filters=[
9             {
10                 'Name': 'tag:Name',
11                 'Values': ['prod']
12             }
13         ]
14     )
15     for volume in volumes['Volumes']:
16         new_snapshot = ec2_client.create_snapshot(
17             VolumeId=volume['VolumeId']
18         )
19         print(new_snapshot)
20
21 schedule.every(20).seconds.do(create_volume_snapshots)
22
```

And here we catch the error:

```
6 def create_volume_snapshots(): 1 usage  Astrid Caballero-Pardo *
7     volumes = ec2_client.describe_volumes(
8         Filters=[
9             {
10                 'Name': 'tag:Name',
11                 'Values': ['prod']
12             }
13         ]
14     )
15     for volume in volumes['Volumes']:
16         try:
17             new_snapshot = ec2_client.create_snapshot(
18                 VolumeId=volume['VolumeId']
19             )
20             print(new_snapshot)
21         except:
22             # handle error here  You, Yesterday • Uncommitted cha
23
```